

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Eighth Semester B.E. (Biotechnology) (Credit System) Degree Examinations**

May - June 2023

19BTC209 - OENOLOGY

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **One full** question from **Unit – I** and **Two full** questions from **Unit – II & Unit – III** each.*

Unit – I		Marks	BT*	CO*	PO*
1. a)	Discuss the constituents of mature grape.	10	L*2	1	12
b)	List out and explain the key steps involved in the prefermentation stage of wine making.	10	L2	2	1,12
2. a)	Explain the physiological effect of pruning on viticulture.	10	L2	1	12
b)	Write a note on				
i)	Dejuicer				
ii)	Supraextraction	10	L2	2	12
Unit – II					
3. a)	Explain anaerobic and aerobic fermentation processes adopted during wine production.	10	L2	3	12
b)	Identify both the advantages and disadvantages associated with the use of oak cooperage in wine production.	10	L2	4	1
4. a)	Using a diagram, demonstrate the construction of a fermenter with a punching down system and describe how this system assists in the fermentation process.	10	L2	3	12
b)	What are some common defects that can occur with cork, and how do they affect the quality of wine?	10	L2	4	1
5. a)	What are the various components that can contribute to instability in wine and explain the techniques that can be used to achieve stability?	10	L2	3	12
b)	Explain various factors that can contribute to wine spoilage.	10	L2	4	1
Unit – III					
6. a)	Discuss the need for laboratory setup at wineries.	10	L2	5	1
b)	What are the various techniques and procedures involved in accurately measuring L-malic acid levels in wine, and how do these methods work?	10	L2	5	1
7. a)	Explain the procedure to determine the insoluble solids present in red wine.	10	L2	5	1
b)	List out and explain visual characterization of red wine.	10	L2	5	1
8. a)	How can CO ₂ levels in a white wine be measured and what steps are involved in this process?	10	L2	5	1
b)	How will you measure the concentration of different metals present in the wine? Explain.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
